SONI KUMARI

Civil & Design Engineer · Jamshedpur, India

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PROFILE

A structurally-grounded Civil and Design Engineer with a career spanning multi-disciplinary environments at SPD Engineering & Consultancy, Energy Design Works, and Amar Found. I specialise in structural layout design, solar PV systems engineering, and green building compliance. Currently at Manupama Technologies Pvt. Ltd., Jamshedpur, I conduct structural and water audits, perform Non-Destructive Testing (NDT), and support IGBC certification — ensuring that quality, safety, and sustainability standards are practically integrated into every project I undertake.

EXPERIENCE

Asst. Structure Engineer · Manupama Technologies Pvt. Ltd., Jamshedpur, India
December 2024 – Present

- Conducted Structural and Water Audits alongside Non-Destructive Testing (NDT), authoring detailed technical compliance and engineering integrity reports for industrial sites.
- Sustainability & Certification: spearheaded documentation and compliance processes to achieve IGBC (Indian Green Building Council) Certification for project sites.
- Engineering Design: developed precise 2D drafting and 3D modelling designs to support facility modifications, audit recommendations, and spatial planning.

Civil Engineer · Amar Found, Surat, Gujarat
April 2024 – December 2024

- Designed reinforced concrete diaphragm walls for deep excavation retention, basement construction, and cut-and-cover applications.
- Produced diaphragm wall panel layout drawings including panel sequencing, guide wall details, and reinforcement cage schedules.
- Coordinated diaphragm wall toe levels with geotechnical investigation data and analysed lateral earth pressure, hydrostatic loads, and surcharge effects using Praxis / WALLAP / CADS.

CAD Engineer · Energy Design Works, Jamshedpur
February 2023 – April 2024

- Designed preliminary layouts and final permit packages based on the client's needs and building property specifications.
- Led project calculations, component sizing and BOM preparation for solar PV systems.
- Designed roof-top and ground-mounted solar arrays in AutoCAD, with accurate system sizing calculations.
- Created Electrical Single-Line Diagrams, panel schedules and connection diagrams for solar electric systems.
- Worked across functions with project management and installation teams to meet clients' needs and internal deadlines.

Junior Engineer · SPD Engineering & Consultancy, Jamshedpur
April 2022 – December 2022

- Plan with furniture layout
- Column layout plan
- Tie beam layout plan
- Centreline plan
- Footing layout plan
- Electrical layout plan
- Plumbing layout plan

INTERNSHIP

Junior Engineer (Intern) · Public Works Department, Building Division, Dhanbad
September 2019 – November 2019

- Regular construction site visits to observe on-ground civil engineering processes and project execution workflows.
- Practical knowledge of building materials — their properties, specifications, grades, and application in structural and finishing works.
- Daily site inspections to monitor the quality of incoming materials and ongoing construction, ensuring adherence to prescribed standards.

EDUCATION

Bachelor of Engineering in Civil Engineering
B.A College of Engineering and Technology (JUT)
Lateral entry: admitted directly to 2nd year in 2021 on Diploma merit
2020 – 2024 · 7.40 CGPA

Diploma in Architecture
Govt. Women's Polytechnic, Jamshedpur (JUT)
2017 – 2020 · 6.69 CGPA

SKILLS

Software
AutoCAD (Proficient), STAAD Pro, MS Office Suite (Word, Excel, PowerPoint)

Structural & Civil Design
AutoCAD drafting, structural layout planning (column, tie beam, centreline, footing), foundation design (pile & diaphragm), anchor report preparation, site inspection & verification

Architectural Drawing
Furniture, electrical and plumbing layout plans for residential and commercial permit packages

Vastu Shastra & Interior Design
Vastu consultation and interior design study

Solar PV Engineering
Rooftop and ground-mounted array design, system sizing, BOM, Single-Line Diagrams, panel schedules and connection diagrams

Audit & Quality
Structural audits, water audits, Non-Destructive Testing (NDT), audit report preparation

Sustainability & Certification
IGBC certification support, green building compliance documentation

PROJECT

Structural Stability & Safety Assessment of Industrial/Commercial Structures

- Comprehensive structural health monitoring, structural auditing, and safety compliance reporting for industrial building sites.
- Non-Destructive Testing (NDT) including Rebound Hammer and Ultrasonic Pulse Velocity (UPV) tests to evaluate concrete compressive strength and integrity.
- Detailed Water Audits and moisture ingress analysis to prevent structural degradation.
- Technical audit reports outlining structural deficiencies, load-bearing capacities, and retrofitting recommendations to meet industrial safety standards.

PROJECT	DRAWN BY	SCALE	SHEET
Soni Kumari	S.K.	N.T.S.	R-001